AllPath Trading Agent

A self-hosted agent that **plans with LLMs, trades with deterministic code, and improves itself every night** · github.com/dukesky/allpath-trading-agent · MIT · 2,368 tests

Options Alpha Agents
Alpaca AI Trading Agents
Hackathon

SYSTEM ARCHITECTURE

① THINK — MULTI-TIER LLM AGENT

You ↔ Agent
web chat · Telegram · CLI — co-create strategies, approve with one tap, or hand it full autonomy per strategy (notify / confirm / auto)

3 model tiers
sonnet-5 chat & drafting · haiku trigger review · opus-5 reflection & memory — strongest model owns the highest-stakes writes

Layered memory (short → long)
observations & conversations (days) → stock dossiers & strategy notes (weeks) → user profile & distilled lessons (durable) — curated markdown, consolidated nightly, human-auditable

Nightly reflection = self-improvement
after each close: review every thesis vs reality → write lessons → re-arm triggers → revise its own strategies (12 auto-applied this week)

② DECIDE — STRATEGIES AS CODE

Strategy YAML
prose thesis + deterministic rules — whitelist-AST conditions, closed action grammar: `buy $3000` · `sell all` · `buy_call $1500 dte>=5 otm=3%` · `close_options`

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Sentinel — every 30 min, zero LLM cost
evaluates every armed rule in plain code; one-shot triggers; this week the intraday path used
no LLM calls at all

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Risk Gate — non-bypassable
order ≤ $5k · position ≤ 25% · options ≤ 10% equity · daily cap · cash floor — and risk *reduction* is never blocked

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Executor — the only path to a broker
the LLM can only produce intents; every fill, skip & rejection journaled to SQLite

③ TRADE — ALPACA INFRASTRUCTURE

Stocks · alpaca-py
paper TradingClient, socket-timeout hardened, notional orders + fill polling

Options · official Alpaca MCP server
managed `uvx alpaca-mcp-server` subprocess (stdio, auto-respawn, 30s timeouts). Deterministic contract selection — the LLM never touches a strike by hand

```
chain(expiry≥dte) →
strike≈spot×(1±otm)
→ quote → qty=⌊budget/(ask×100)⌋
→ place_option_order → journal →
notify
```

Safety net, always on
15% drawdown breaker (halts new risk, exits keep working) · DTE≤1 expiry sweep · market-hours discipline · honest failure: no order id ⇒ journaled error + page, never a phantom fill

WHAT MAKES IT DIFFERENT

- **LLM thinks nightly; code trades intraday.** Inference goes to research and reflection — execution is deterministic, auditable, outage-proof.
- **It genuinely improves itself.** Night 1 it found a stop→re-entry flaw in all 5 of its own strategies, unprompted; mid-week it wrote salvage rules that recovered **~$1,165** of decaying premium; its last report critiques its own stops.
- **Autonomy is earned, bounded, and revocable.** Per-strategy tiers; revisions pass byte-exact staleness + version checks; a freeze bars the agent from ever raising its own authorization.
- **Everything is explainable.** Strategies are readable YAML; every decision has a journal row; every nightly report ships a full tool-call transcript.

THE LIVE RUN — HONEST SCOREBOARD

- Fresh **$100k** paper account · one kickoff chat · then **5 days hands-off** (0 human trading decisions)
- **17 rule-driven fills** — 4 stock legs + 5 option legs in, every exit by stop / salvage / expiry sweep
- **12 nightly self-revisions** auto-applied through the guarded pipeline
- **–4.9%** ($95,138): stock book ≈ flat (–$180); the 5-DTE options overlay paid –$4.7k tuition — **diagnosed in writing by the agent itself on night one**, then corrected all week

Judges' shortcut: the **Reports page** replays every nightly reflection with its full tool-call transcript — the audit trail *is* the demo.